KAIZEN MATHS

Lesson starter

Integration of Algebraic Fractions

Simplify Before Integrating

Name _____

S1

Cancel the common factor before integrating.

Find $\int \frac{(x+2)(6x+6)}{x+2}\,dx$.

S2

Use polynomial division before integrating.

Find $\int \frac{5x^2+23x+13}{x+4}\,dx$.

S3

Use polynomial division before integrating.

Find $\int \frac{3x^2-2x}{x-1}\,dx$.

S4

Cancel the common factor before integrating.

Find $\int \frac{(x+3)(3x-6)}{x+3}\,dx$.

Show your working

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ALG-31E3-L2 · S

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Exit ticket

Integration of Algebraic Fractions

Simplify Before Integrating

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E1

Use polynomial division before integrating.

Find $\int \frac{5x^2-11x+10}{x-2}\,dx$.

E2

Use polynomial division before integrating.

Find $\int \frac{x^2-x-32}{x+5}\,dx$.

E3

Cancel the common factor before integrating.

Find $\int \frac{(x+2)(6x-5)}{x+2}\,dx$.

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ALG-31E3-L2 · E

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ALG-31E3-L2 · E

Show your working

Teacher answers & working

Integration of Algebraic Fractions · Simplify Before Integrating

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S1

Cancel the common factor before integrating.

Find $\int \frac{(x+2)(6x+6)}{x+2}\,dx$.

Answer:

$3x^2 + 6x + C$

Working:

The numerator and denominator share a common factor of (x + 2).

$\frac{(x+2)(6x+6)}{x+2} = 6x + 6.$

Now integrate the simplified expression.

$\int (6x + 6)\,dx = 3x^2 + 6x + C.$

S2

Use polynomial division before integrating.

Find $\int \frac{5x^2+23x+13}{x+4}\,dx$.

Answer:

$\frac{5}{2}x^2 + 3x + \ln|x + 4| + C$

Working:

Use division to rewrite the improper fraction.

$5x^2 + 23x + 13 = (x + 4)(5x + 3) + 1.$

So $\frac{5x^2+23x+13}{x+4} = 5x + 3 + \frac{1}{x+4}$.

Integrate each part: $\int (5x + 3)\,dx = \frac{5}{2}x^2 + 3x$.

Also $\int \frac{1}{x+4}\,dx = 1\ln|x + 4|$.

Final answer: $\frac{5}{2}x^2 + 3x + \ln|x + 4| + C$.

S3

Use polynomial division before integrating.

Find $\int \frac{3x^2-2x}{x-1}\,dx$.

Answer:

$\frac{3}{2}x^2 + x + \ln|x - 1| + C$

Working:

Use division to rewrite the improper fraction.

$3x^2 - 2x = (x - 1)(3x + 1) + 1.$

So $\frac{3x^2-2x}{x-1} = 3x + 1 + \frac{1}{x-1}$.

Integrate each part: $\int (3x + 1)\,dx = \frac{3}{2}x^2 + x$.

Also $\int \frac{1}{x-1}\,dx = 1\ln|x - 1|$.

Final answer: $\frac{3}{2}x^2 + x + \ln|x - 1| + C$.

Teacher answers & working

Integration of Algebraic Fractions · Simplify Before Integrating

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S4

Cancel the common factor before integrating.

Find $\int \frac{(x+3)(3x-6)}{x+3}\,dx$.

Answer:

$\frac{3}{2}x^2 - 6x + C$

Working:

The numerator and denominator share a common factor of (x + 3).

$\frac{(x+3)(3x-6)}{x+3} = 3x - 6$.

Now integrate the simplified expression.

$\int (3x - 6)\,dx = \frac{3}{2}x^2 - 6x + C$.

E1

Use polynomial division before integrating.

Find $\int \frac{5x^2-11x+10}{x-2}\,dx$.

Answer:

$\frac{5}{2}x^2 - x + 8\ln|x - 2| + C$

Working:

Use division to rewrite the improper fraction.

$5x^2 - 11x + 10 = (x - 2)(5x - 1) + 8$.

So $\frac{5x^2-11x+10}{x-2} = 5x - 1 + \frac{8}{x-2}$.

Integrate each part: $\int (5x - 1)\,dx = \frac{5}{2}x^2 - x$.

Also $\int \frac{8}{x-2}\,dx = 8\ln|x - 2|$.

Final answer: $\frac{5}{2}x^2 - x + 8\ln|x - 2| + C$.

E2

Use polynomial division before integrating.

Find $\int \frac{x^2-x-32}{x+5}\,dx$.

Answer:

$\frac{1}{2}x^2 - 6x - 2\ln|x + 5| + C$

Working:

Use division to rewrite the improper fraction.

$x^2 - x - 32 = (x + 5)(x - 6) - 2$.

So $\frac{x^2-x-32}{x+5} = x - 6 + \frac{-2}{x+5}$.

Integrate each part: $\int (x - 6)\,dx = \frac{1}{2}x^2 - 6x$.

Also $\int \frac{-2}{x+5}\,dx = -2\ln|x + 5|$.

Final answer: $\frac{1}{2}x^2 - 6x - 2\ln|x + 5| + C$.

Teacher answers & working

Integration of Algebraic Fractions · Simplify Before Integrating

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E3

Cancel the common factor before integrating.

Find $\int \frac{(x+2)(6x-5)}{x+2}\,dx$.

Answer:

$3x^2 - 5x + C$

Working:

The numerator and denominator share a common factor of (x + 2).

$\frac{(x+2)(6x-5)}{x+2} = 6x - 5$.

Now integrate the simplified expression.

$\int (6x - 5)\,dx = 3x^2 - 5x + C$.